Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of "CUBIC Diffusion" Equation

Proof. Similar to the derived equation on changes in lipid concentration, the CUBIC Diffusion function is also based on the Continuity Equation for Mass:

$$\frac{\partial C}{\partial t} + \nabla \cdot N_i = 0$$

Molar flux (N_i), which is defined as

$$N_i = -D_i \nabla C$$

can be substituted into the Continuity Equation, yielding

$$\frac{\partial C}{\partial t} = D \nabla^2 C.$$

The convective flux component of the Convection-Diffusion equation and Darcy's Law are both coupled into the intermediate equation:

$$\frac{\partial C}{\partial t} = D \nabla^2 C + \frac{k}{\mu} \nabla \cdot (C \nabla p)$$

Diffusivity (D) is substituted with Brownian Motion's Formula, which also includes porosity and tortuosity as a component. To simplify tortuosity, Bruggeman's Model was substituted, resulting in

$$D = \frac{(x - x_0)^2}{4(t - t_0)} \varepsilon_0^{\frac{3}{2}}.$$

Simplification of the intermediate equation yields the complete equation:

$$\boxed{\frac{\partial C}{\partial t} = \frac{(x - x_0^2)}{4(t - t_0)} \varepsilon^{(\frac{3}{2}+n)} \nabla^2 C + \frac{k}{\mu} (\nabla \cdot (C \nabla p))}.$$